

# **Chapter 7 Analysis and Design of Linear Discrete-Time System (Sampled-data System)**

## **7.1 Introduction**

## **7.2 The Sampling Process and Sampling Theorem**

## **7.3 Signal Recovery and Zero-Order Hold**

## **7.4 Z-Transform and Inverse Z Transform**

## **7.5 Mathematical Models of Discrete-Time Systems**

## **7.6 Performance Analysis of Discrete-Time Systems**

## **7.7 Digital Control Design for Discrete-Time Systems**

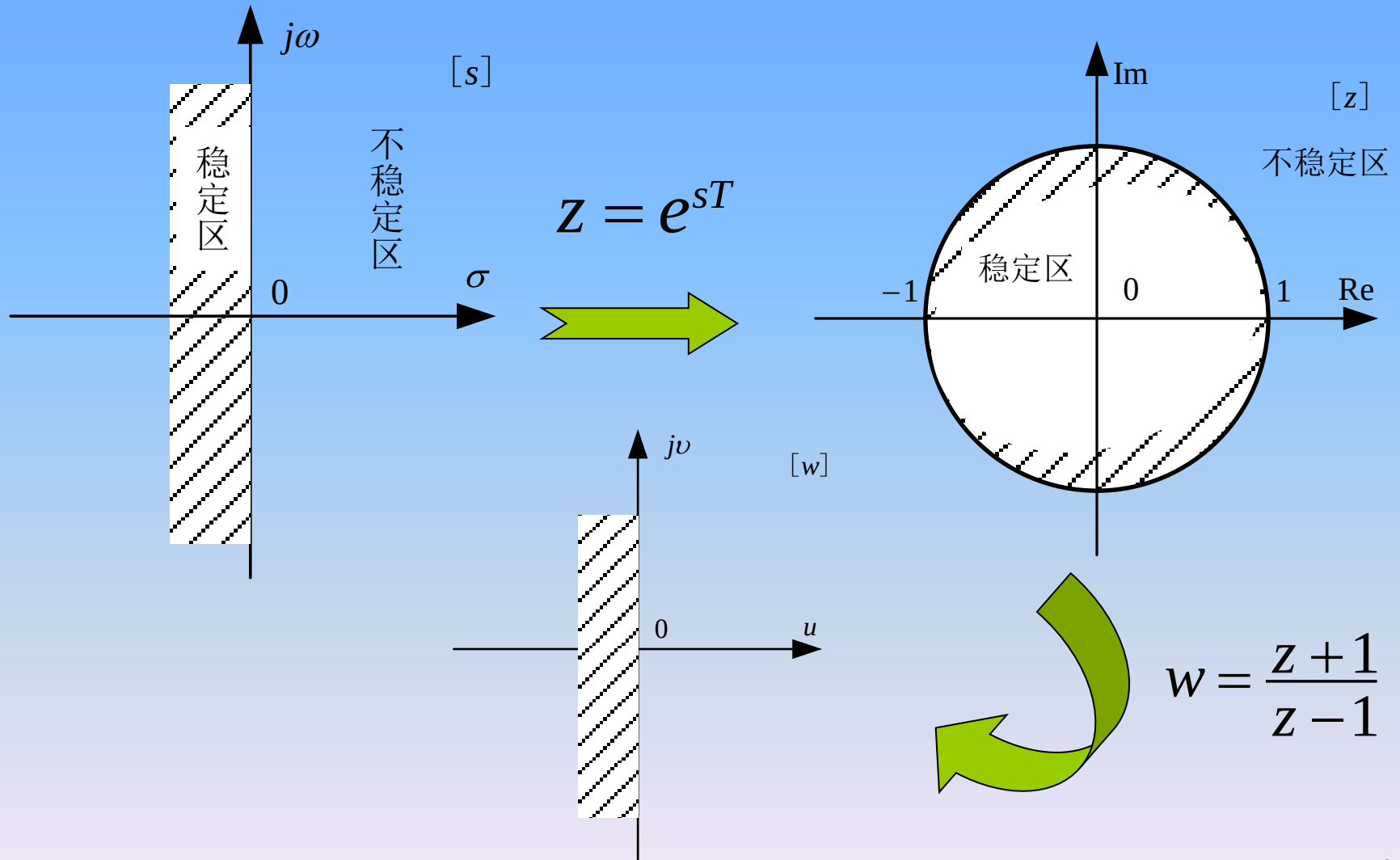
## **s-Domain to z-Domain Mapping**

### **Necessary and Sufficient Condition for Stability of Linear Discrete-Time Systems**

**— All poles of  $\Phi(z)$  lie in the unit circle of z plane**

### **Routh criterion in w domain (Generalized Routh Criterion)**

**we've learned three methods to determine the stability of a discrete-time systems.**



## 7.6 Performance Analysis of Discrete-Time Systems

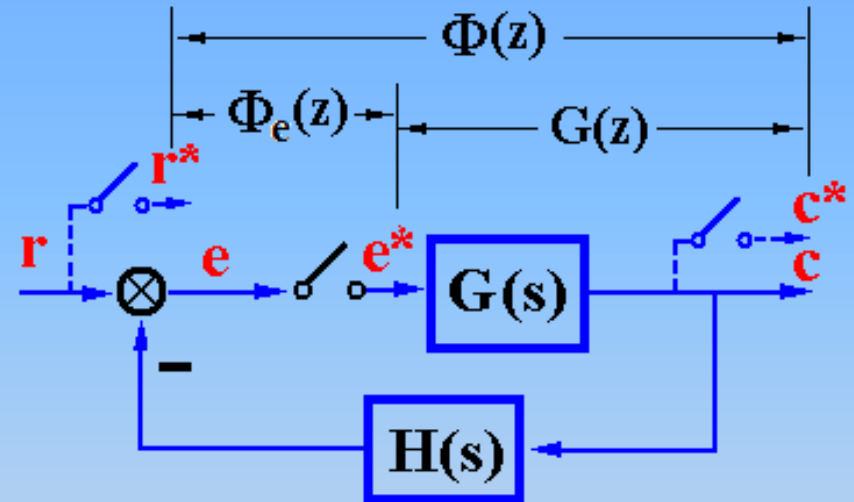
- Stability
- **Dynamic Performance**
- Steady-state Errors

## 7.6.2 Dynamic Performance Analysis of Discrete-Time Systems

### 1. General algorithm to obtain the dynamic performance

(1) Obtain the impulse transfer function

$$\text{Let } \begin{cases} GH(z) = Z[G(s)H(s)] \\ \Phi(z) = \frac{G(z)}{1 + GH(z)} = \frac{M(z)}{D(z)} \end{cases}$$

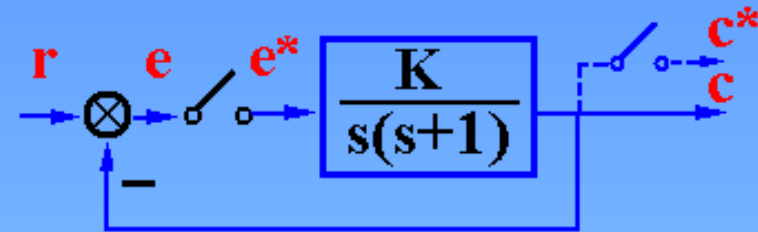


$$\begin{aligned} (2) \text{ Obtain } C(z) &= \Phi(z)R(z) = \frac{M(z)}{D(z)} \cdot \frac{z}{z-1} \\ &= c(0) + c(T)z^{-1} + c(2T)z^{-2} + \dots \end{aligned}$$

$$(3) \quad c^*(t) = c(0)\delta(t) + c(T)\delta(t-T) + c(2T)\delta(t-2T) + \dots$$

(4) Determine the specifications  $\sigma\%$ ,  $t_s$ .

**Example 1** Consider the system shown in the figure,  $T=K=1$ . Obtain the dynamic specifications. ( $\sigma\%$ ,  $t_s$ ).



**Solution.** 
$$G(z) = Z \left[ \frac{K}{s(s+1)} \right] = \frac{K(1-e^{-T})z}{(z-1)(z-e^{-T})}$$

$$\stackrel{K=T=1}{=} \frac{0.632z}{(z-1)(z-0.368)}$$

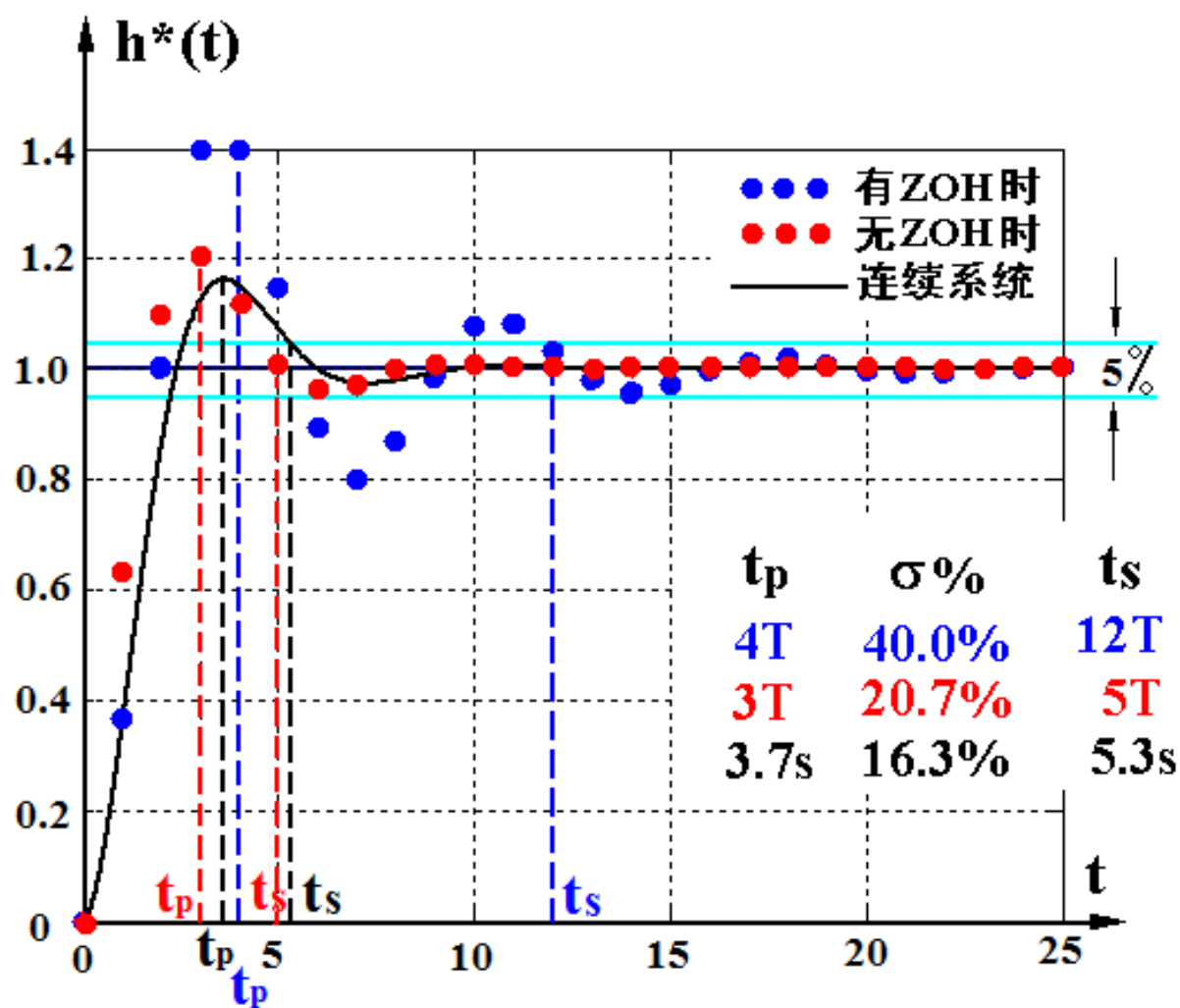
$$\Phi(z) = \frac{G(z)}{1+G(z)} = \frac{0.632z}{z^2 - 0.736z + 0.368}$$

$$c(\infty T) = \lim_{z \rightarrow 1} (z-1) \cdot \Phi(z) \cdot \frac{z}{z-1} = 1$$

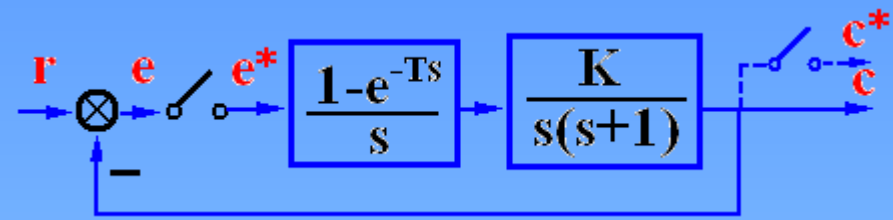
$$C(z) = \Phi(z) \cdot \frac{z}{z-1} = \frac{0.632z^2}{z^3 - 1.736z^2 + 1.104z - 0.368}$$

Obtain the unit step response series  $h(k)$  by long division method.

$$\begin{aligned} h(0) &= 0 \\ h(1) &= 0.632 \\ h(2) &= 1.097 \\ h(3) &= 1.207 \\ h(4) &= 1.117 \\ h(5) &= 1.014 \\ h(6) &= 0.964 \\ h(7) &= 0.970 \\ h(8) &= 0.991 \\ h(9) &= 1.004 \\ h(10) &= 1.007 \\ h(11) &= 1.003 \\ h(12) &= 1.000 \\ &\vdots \end{aligned} \left\{ \begin{array}{l} t_p = 3T \\ \sigma\% = 20.7\% \\ t_s = 5T \end{array} \right.$$



**Example 1** Consider the system shown in the figure,  $T=K=1$ . Obtain the dynamic specifications. ( $\sigma\%$ ,  $t_s$ ).



Solution. 
$$G(z) = K \frac{z-1}{z} Z \left[ \frac{1}{s^2(s+1)} \right]$$

$$= K \frac{(T-1+e^{-T})z + (1-e^{-T}-Te^{-T})}{(z-1)(z-e^{-T})}$$

$$\stackrel{K=T=1}{=} \frac{0.368z + 0.264}{(z-1)(z-0.368)}$$

$$\Phi(z) = \frac{G(z)}{1+G(z)} = \frac{0.368z + 0.264}{z^2 - z + 0.632}$$

$$c(\infty T) = \lim_{z \rightarrow 1} (z-1) \cdot \Phi(z) \cdot \frac{z}{z-1} = 1$$

$$C(z) = \Phi(z) \cdot \frac{z}{z-1} = \frac{(0.368z + 0.264)z}{z^3 - 2z^2 + 1.632z - 0.632}$$

$$h(0)=0$$

$$h(1)=0.3679$$

$$h(2)=1.0000$$

$$h(3)=1.3996$$

$$h(4)=1.3996$$

$$h(5)=1.1470$$

$$h(6)=0.8944$$

$$h(7)=0.8015$$

$$h(8)=0.8682$$

$$h(9)=0.9937$$

$$h(10)=1.0770$$

$$h(11)=1.0810$$

$$h(12)=1.0323$$

$$h(13)=0.9811$$

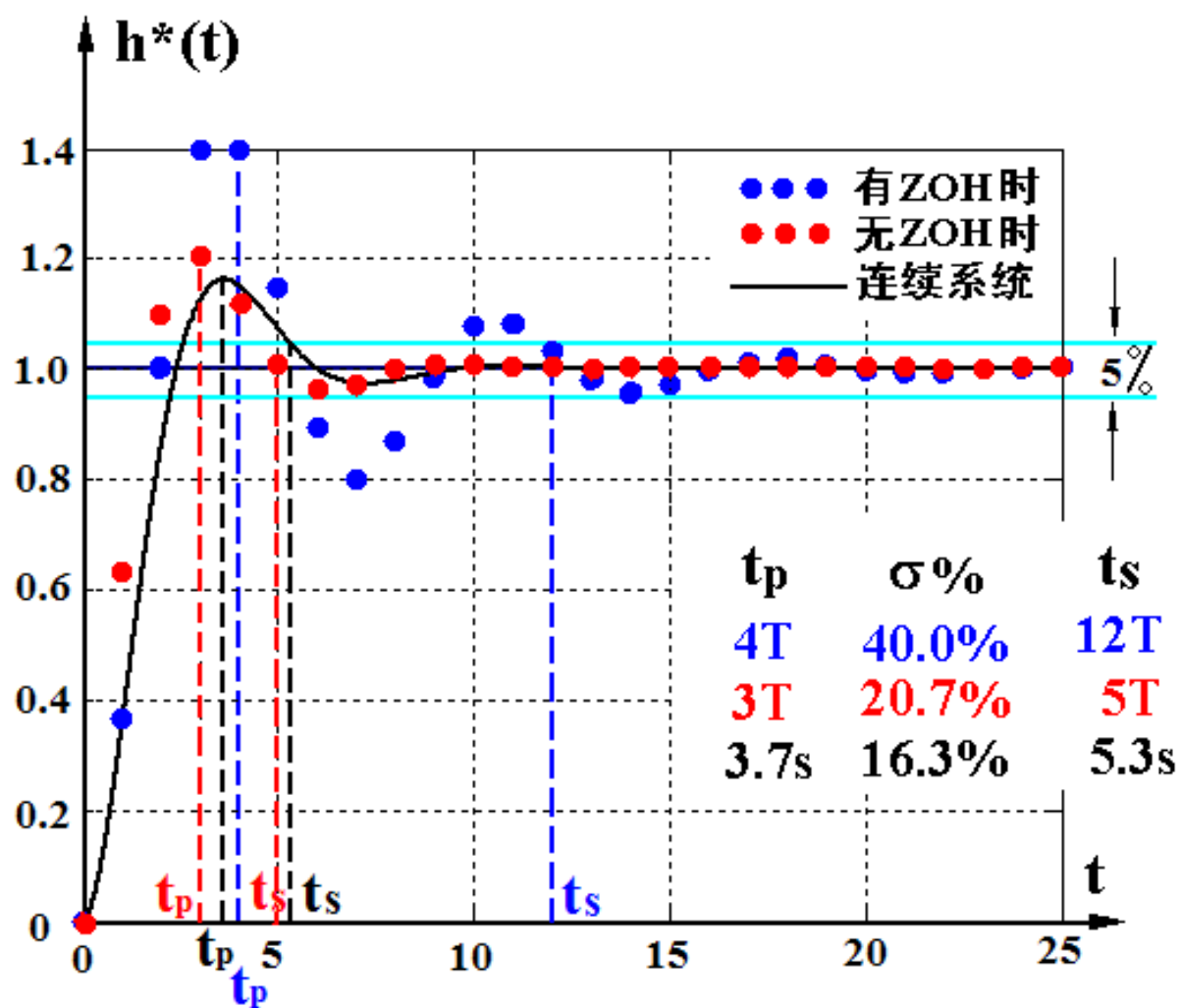
$$h(14)=0.9607$$

$\vdots$

$$\left\{ \begin{array}{l} t_p = 4T \\ \sigma\% = 40\% \end{array} \right.$$

$$t_s = 12T$$





## 2. Relationship between dynamic response and closed-loop poles

$$\Phi(z) = \frac{M(z)}{D(z)} = \frac{b_m \prod_{i=1}^m (z - z_i)}{a_n \prod_{k=1}^n (z - p_k)} \quad m \leq n$$

$$\begin{aligned} C(z) &= \Phi(z)R(z) = \frac{M(z)}{D(z)} \cdot \frac{z}{z-1} \\ &= \frac{M(1)}{D(1)} \cdot \frac{z}{z-1} + \sum_{k=1}^n \frac{c_k z}{z - p_k} \end{aligned}$$

### (1) Single closed-loop poles on the real axis

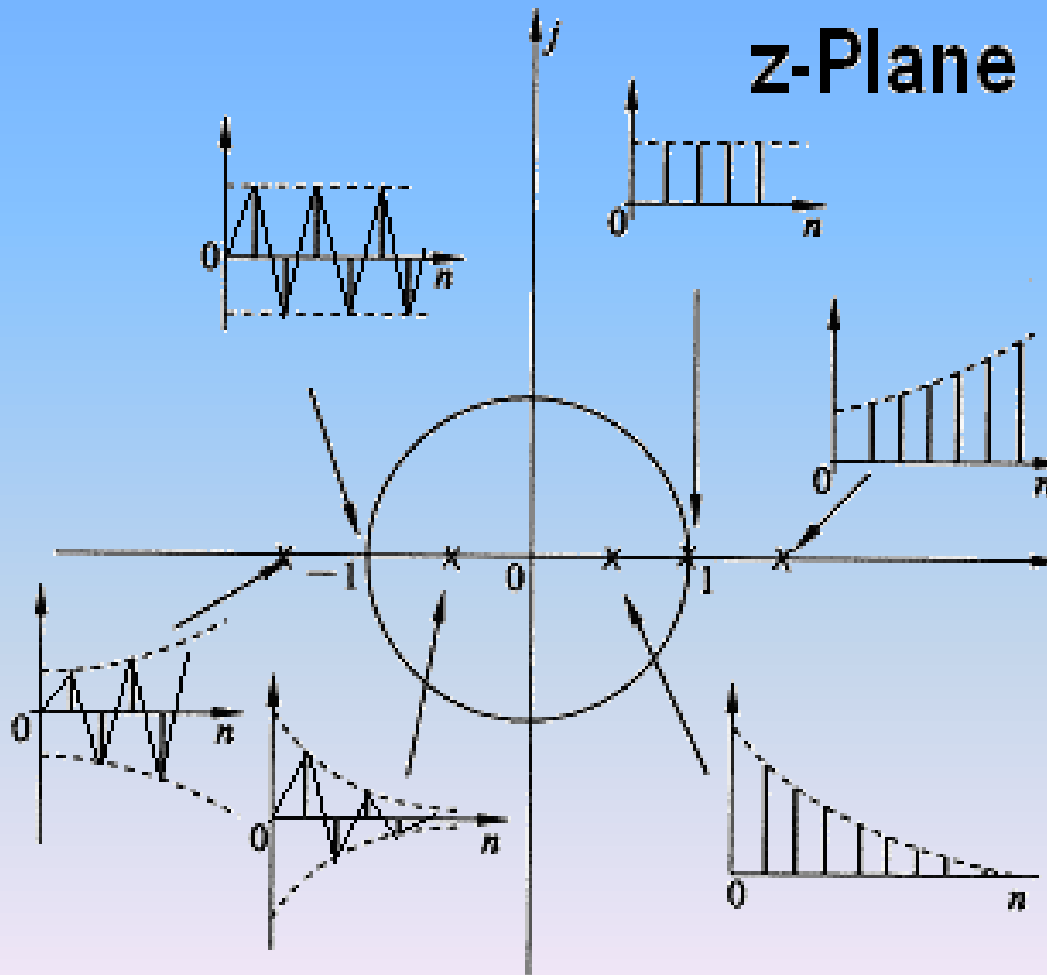
$$c_k^*(t) = Z^{-1} \left[ \frac{c_k z}{z - p_k} \right] \quad k = 1, 2, \dots, n$$

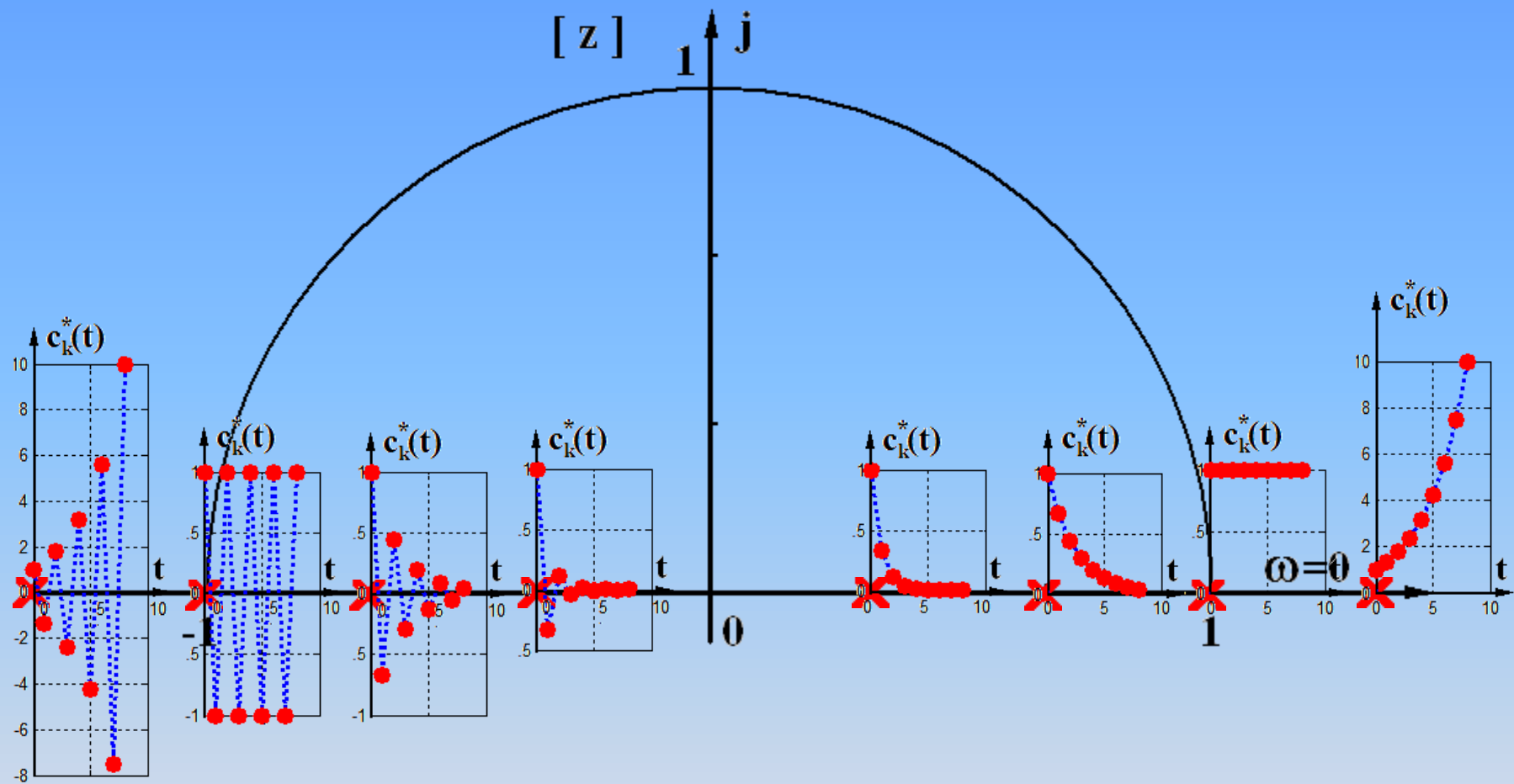
$$c_k(nT) = c_k p_k^n \quad k = 1, 2, \dots, n$$

$$c_k(nT) = c_k p_k^n \quad k = 1, 2, \dots, n$$

$$p_k > 0: \quad p_k > 1 \quad p_k = 1 \quad p_k < 1$$

$$p_k < 0:$$





## (2) Closed-loop Complex conjugate poles

$$p_k = |p_k| e^{j\theta_k} \quad \bar{p}_k = |p_k| e^{-j\theta_k}$$

$$c_{k,k}^*(k) = Z^{-1} \left[ \frac{c_k z}{z - p_k} + \frac{\bar{c}_k z}{z - \bar{p}_k} \right]$$

$$\begin{cases} a = \frac{1}{T} \ln |p_k| \\ \omega = \frac{\theta_k}{T} \\ 0 < \theta_k < \pi \end{cases}$$

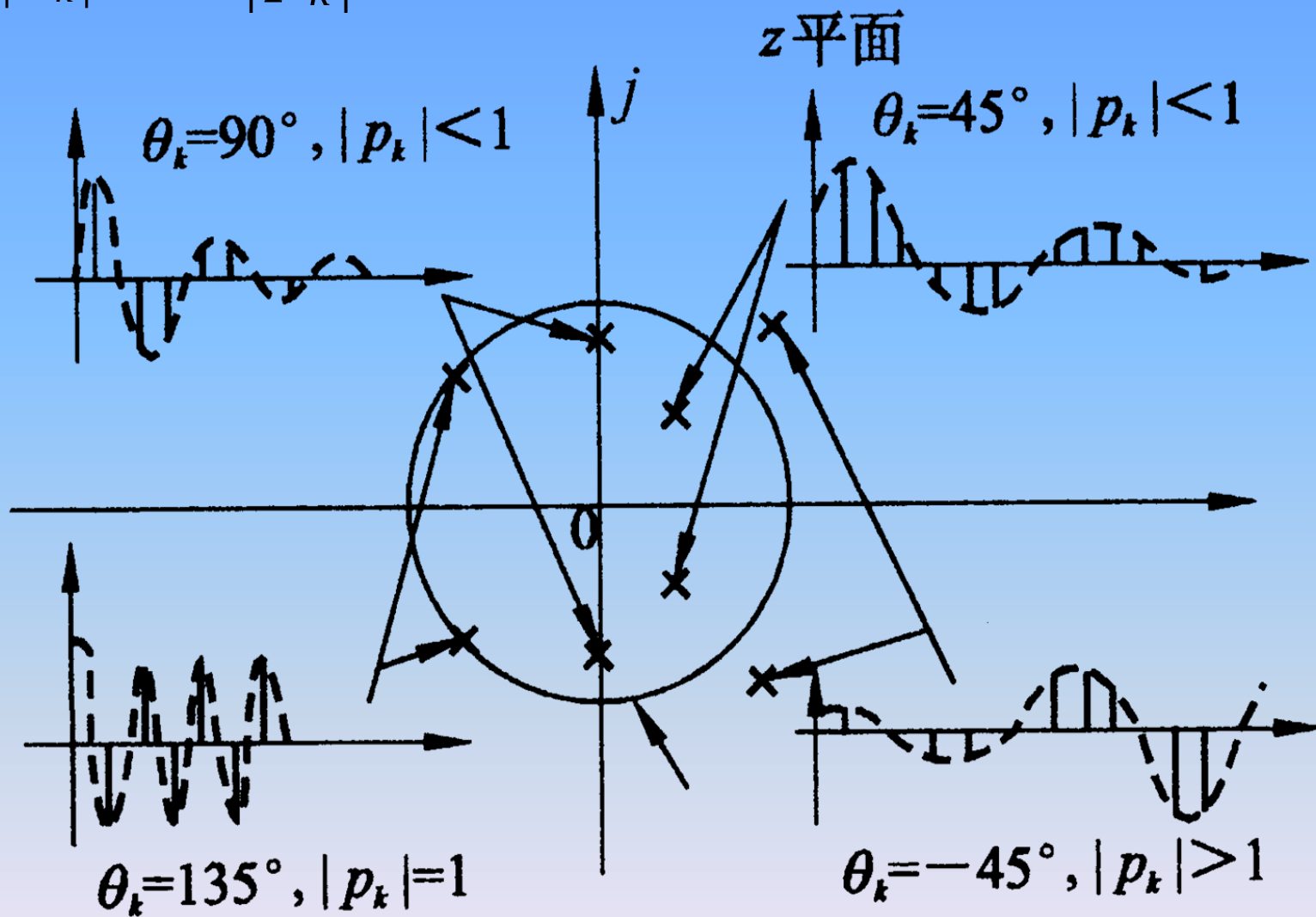
$$c_{k,k}(nT) = c_k p_k^n + \bar{c}_k \bar{p}_k^n$$

$$= c_k e^{a_k nT} + \bar{c}_k e^{\bar{a}_k nT}$$

$$= |c_k| e^{j\varphi_k} e^{(a+j\omega)nT} + |c_k| e^{-j\varphi_k} e^{(a-j\omega)nT}$$

$$= 2|c_k| e^{anT} \cos(n\omega T + \varphi_k)$$

$$|p_k| < 1, \quad |p_k| > 1$$



## 7.6 Performance Analysis of Discrete-Time Systems

- **Stability**
- **Dynamic Performance**
- **Steady-state Errors**

## 7.6.2 Analysis of discrete-time dynamic performance

(1) General method

$$\left\{ \begin{array}{l} G(z) \rightarrow \Phi(z) \longrightarrow C(z) = \sum_{n=0}^{\infty} c(nT)z^{-n} \\ c^*(t) = \sum_{n=0}^{\infty} c(nT)\delta(t - nT) \longrightarrow \text{Obtain } s\%, t_s \text{ by definition} \end{array} \right.$$

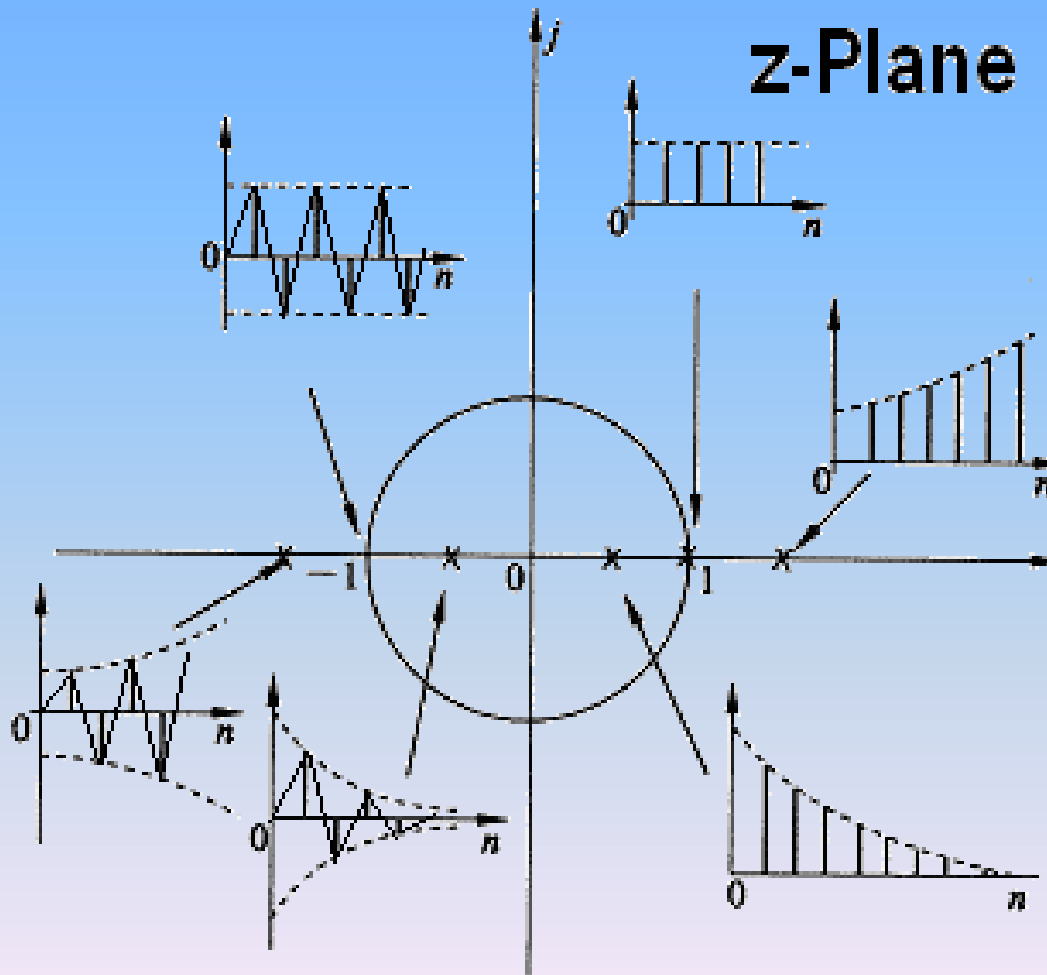
(2) Closed-loop poles  $p_k \longrightarrow$  Response  $c_k(nT) = C_k p_k^n$



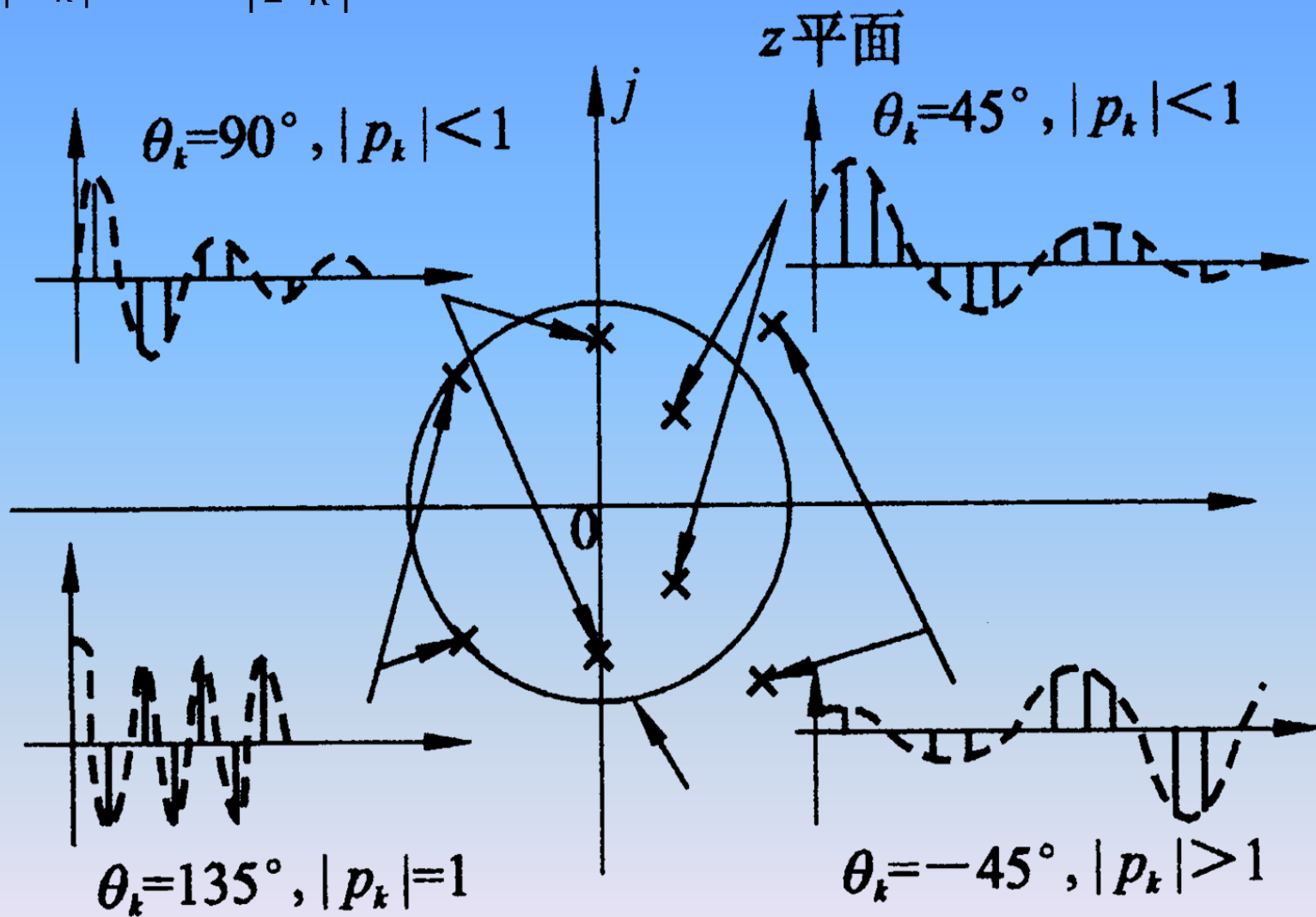
$$c_k(nT) = c_k p_k^n \quad k = 1, 2, \dots, n$$

$$p_k > 0: \quad p_k > 1 \quad p_k = 1 \quad p_k < 1$$

$$p_k < 0:$$



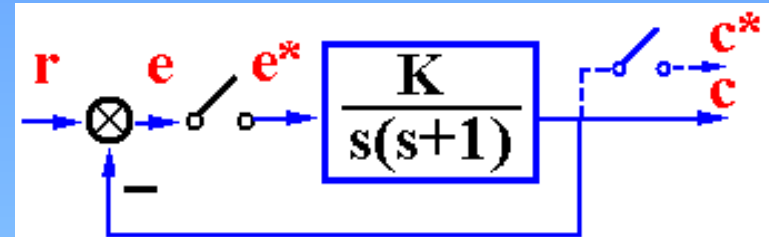
$$|p_k| < 1, \quad |p_k| > 1$$



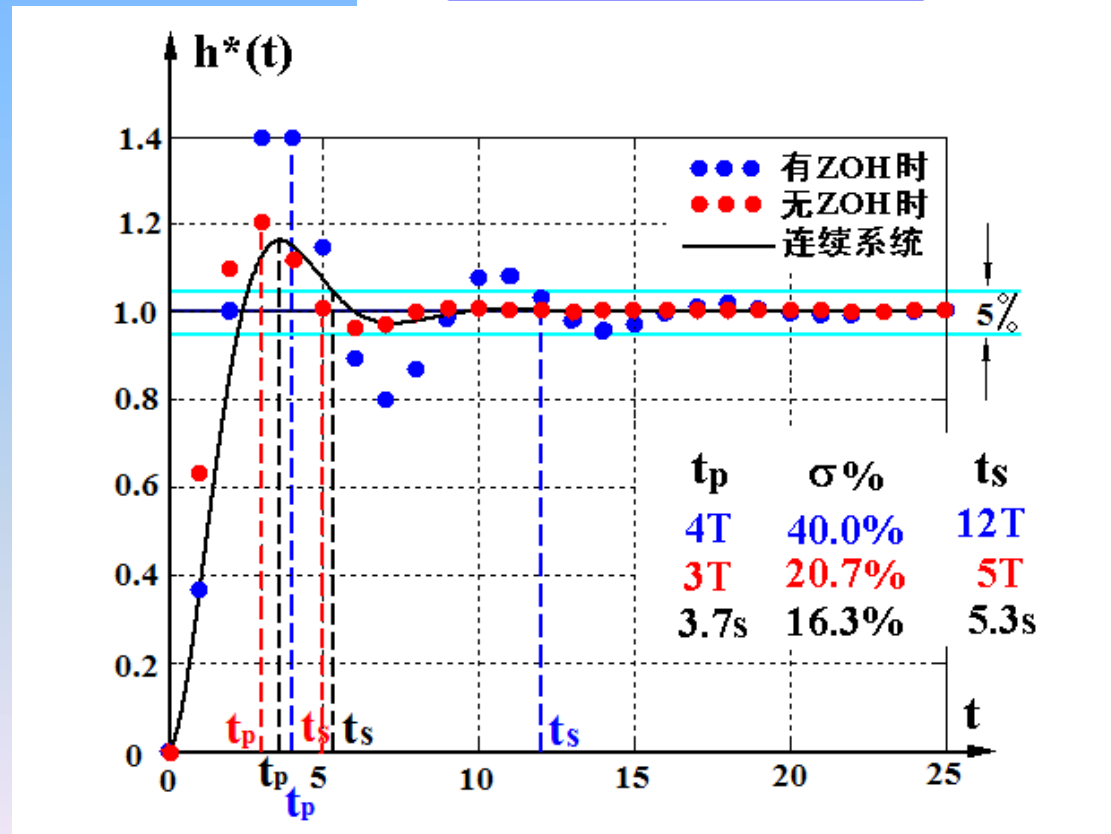
## 7.6.3 Steady-state error

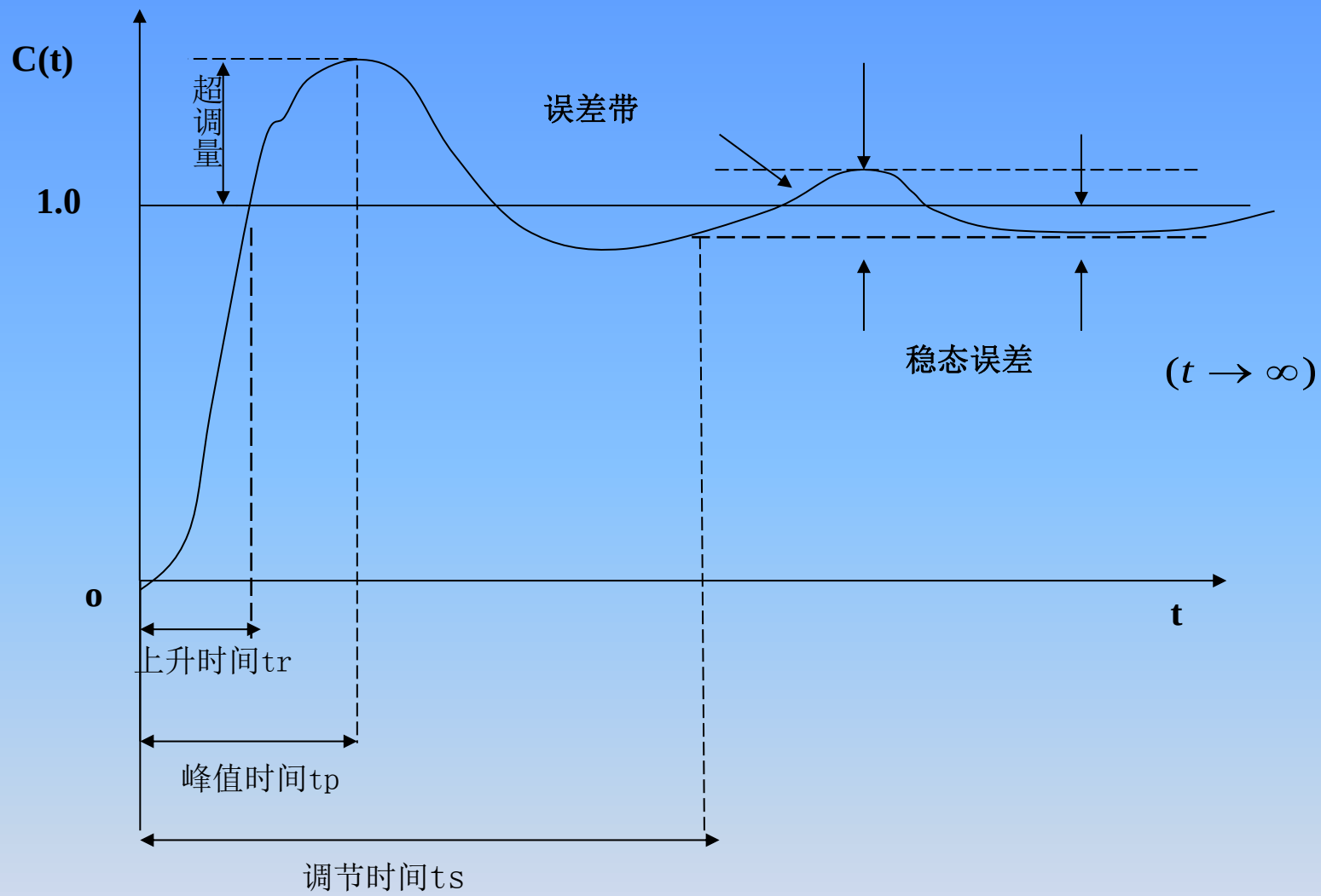
### 1. General method to obtain steady-state error

**Example** Consider the system shown in the figure,  $T=K=1$ . Obtain the dynamic specifications. ( $\sigma\%$ ,  $t_s$ ).



$h(0)=0$   
 $h(1)=0.632$   
 $h(2)=1.097$   
 $h(3)=1.207$   
 $h(4)=1.117$   
 $h(5)=1.014$   
 $h(6)=0.964$   
 $h(7)=0.970$   
 $h(8)=0.991$   
 $h(9)=1.004$   
 $h(10)=1.007$   
 $h(11)=1.003$   
 $h(12)=1.000$   
 $\vdots$





控制系统性能指标

## 2. Using final value theorem to obtain steady-state error

Let  $\begin{cases} GH(z) = Z[G(s)H(s)] = \frac{1}{(z-1)^v} GH_0(z) \\ \lim_{z \rightarrow 1} GH_0(z) = K \end{cases}$  **v: System type**

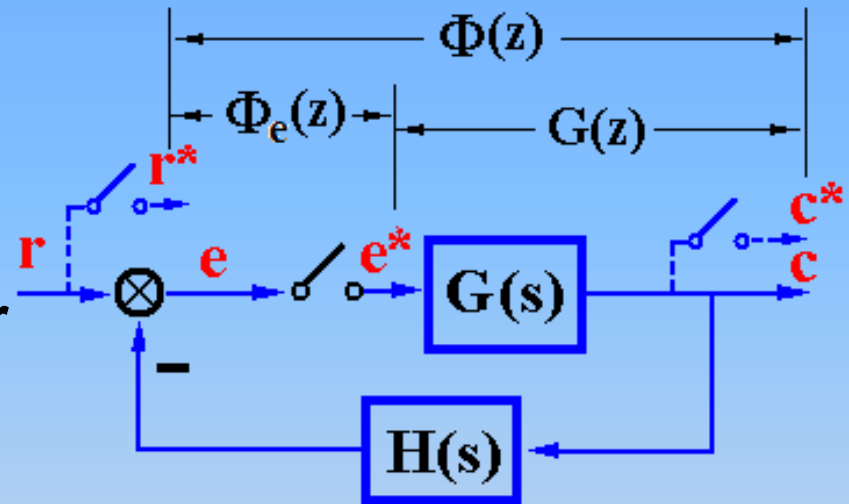
Algorithm:

- (1) Determine the stability
- (2) Obtain the impulse transfer function from  $E(z)$  to  $R(z)$ .

$$\Phi_e(z) = \frac{E(z)}{R(z)} = \frac{1}{1 + GH(z)}$$

- (3) Obtain  $e(\infty)$  by the final value theorem

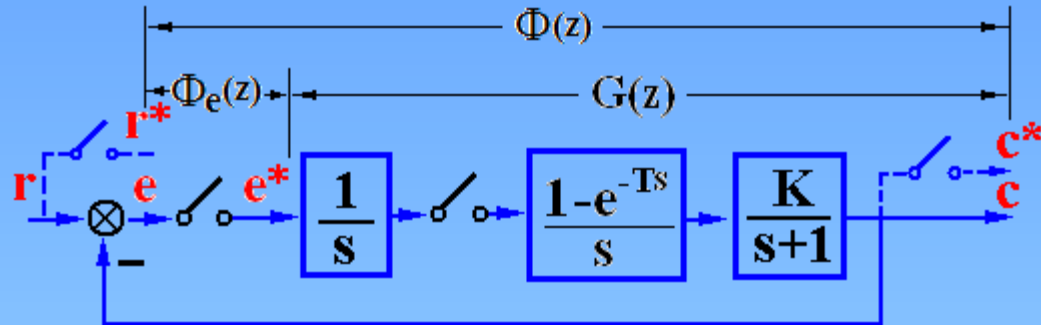
$$e(\infty) = \lim_{z \rightarrow 1} (z-1) \Phi_e(z) R(z) = \lim_{z \rightarrow 1} (z-1) \cdot R(z) \cdot \frac{1}{1 + GH(z)}$$



**Example 1** Consider the discrete system shown in the figure,  $K=2$ ,  $T=1$ ; Obtain  $e(\infty)$  for  $r(t)=1(t)$ ,  $t$ ,  $t^2/2$ .

$$G(z) = Z\left[\frac{1}{s}\right] \cdot Z\left[\frac{1-e^{-Ts}}{s} \cdot \frac{K}{s+1}\right]$$

$$= \frac{K(1-e^{-T})z}{(z-1)(z-e^{-T})} \quad \mathbf{v = 1}$$



$$\Phi_e(z) = \frac{1}{1 + \frac{K(1-e^{-T})z}{(z-1)(z-e^{-T})}} = \frac{(z-1)(z-e^{-T})}{(z-1)(z-e^{-T}) + K(1-e^{-T})z}$$

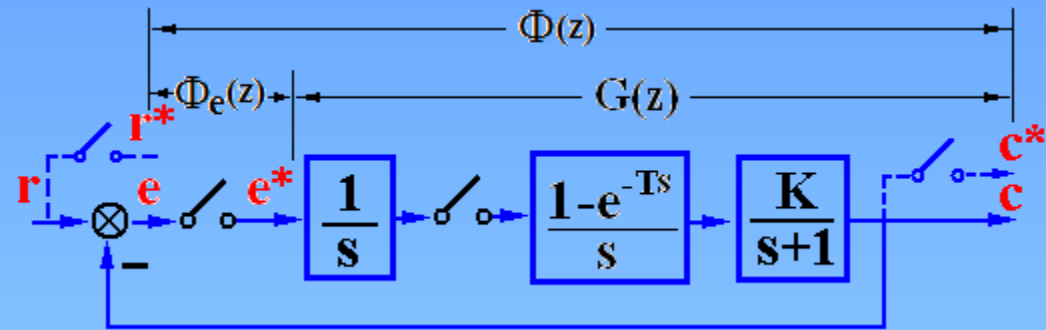
$$D(z) = z^2 + [K(1-e^{-T}) - (1+e^{-T})]z + e^{-T} = 0$$

$$0 < K < \frac{2(1+e^{-T})}{(1-e^{-T})} \quad T=1 = 4.33$$

**Example 1** Consider the discrete system shown in the figure,  $K=2$ ,  $T=1$ ;  
Obtain  $e(\infty)$  for  $r(t)=1(t)$ ,  $t$ ,  $t^2/2$ .

$$0 < K < 4.33$$

$$e(\infty) = \lim_{z \rightarrow 1} (z-1) R(z) \Phi_e(z)$$



$$\Phi_e(z) = \frac{(z-1)(z-e^{-T})}{(z-1)(z-e^{-T}) + K(1-e^{-T})z}$$

$$r_1(t) = 1(t) \quad e_1(\infty) = \lim_{z \rightarrow 1} \cancel{(z-1)} \frac{z}{z-1} \cdot \frac{(z-1)(z-e^{-T})}{(z-1)(z-e^{-T}) + K(1-e^{-T})z} = 0$$

$$r_2(t) = t \quad e_2(\infty) = \lim_{z \rightarrow 1} \cancel{(z-1)} \frac{Tz}{(z-1)^2} \cdot \frac{(z-1)(z-e^{-T})}{(z-1)(z-e^{-T}) + K(1-e^{-T})z} = \frac{T}{K}$$

$$r_3(t) = \frac{t^2}{2} \quad e_3(\infty) = \lim_{z \rightarrow 1} \cancel{(z-1)} \frac{Tz(z+1)}{2(z-1)^3} \cdot \frac{(z-1)(z-e^{-T})}{(z-1)(z-e^{-T}) + K(1-e^{-T})z} = \infty$$

### 3. Static Error Constant Method

shows how  $e(\infty)$  changes with  $r(t)$

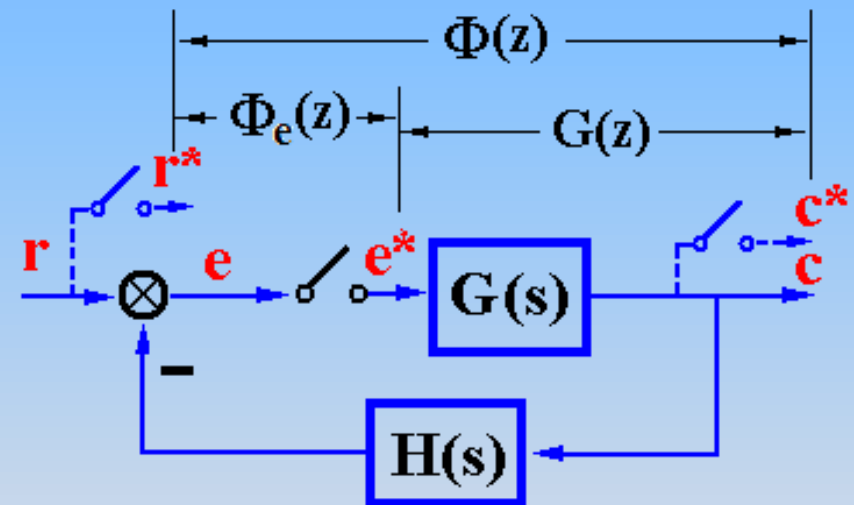
(For stable linear discrete systems subject to  $r(t)$  and sampled at the error signal)

Let  $\begin{cases} GH(z) = Z[G(s)H(s)] = \frac{1}{(z-1)^v} GH_0(z) & \text{v: System type} \\ \lim_{z \rightarrow 1} GH_0(z) = K \end{cases}$

$$\Phi_e(z) = \frac{E(z)}{R(z)} = \frac{1}{1 + GH(z)}$$

$$e(\infty) = \lim_{z \rightarrow 1} (z-1) \Phi_e(z) R(z)$$

$$= \lim_{z \rightarrow 1} (z-1) \cdot R(z) \cdot \frac{1}{1 + GH(z)}$$





$$e(\infty T) = \lim_{z \rightarrow 1} (z-1) \Phi_e(z) R(z) = \lim_{z \rightarrow 1} (z-1) \cdot R(z) \cdot \frac{1}{1+GH(z)}$$

$$r(t) = A \cdot 1(t) \quad e(\infty T) = \lim_{z \rightarrow 1} (z-1) \cdot \frac{Az}{z-1} \cdot \frac{1}{1+GH(z)} = \frac{A}{1 + \lim_{z \rightarrow 1} GH(z)} = \frac{A}{K_p}$$

Static position error constant

$$K_p = 1 + \lim_{z \rightarrow 1} GH(z)$$

$$r(t) = A \cdot t \quad e(\infty T) = \lim_{z \rightarrow 1} (z-1) \cdot \frac{ATz}{(z-1)^2} \cdot \frac{1}{1+GH(z)} = \frac{AT}{\lim_{z \rightarrow 1} (z-1)GH(z)} = \frac{AT}{K_v}$$

Static velocity error constant

$$K_v = \lim_{z \rightarrow 1} (z-1)GH(z)$$

$$r(t) = \frac{A}{2} t^2 \quad e(\infty T) = \lim_{z \rightarrow 1} (z-1) \cdot \frac{AT^2 z(z+1)}{2(z-1)^3} \cdot \frac{1}{1+GH(z)} = \frac{AT^2}{\lim_{z \rightarrow 1} (z-1)^2 GH(z)} = \frac{AT^2}{K_a}$$

Static acceleration error constant

$$K_a = \lim_{z \rightarrow 1} (z-1)^2 GH(z)$$

**Similar to the continuous system, we can divide the discrete-time system as type 0, type I, type II,... according to the numbers of the pole  $z=1$  of the impulse transfer function.**

$$r(t) = A \cdot 1(t)$$

**Static position error constant**

$$K_p = 1 + \lim_{z \rightarrow 1} GH(z)$$

**Type 0:  $K_p = \text{constant}$**

**Type  $\geq 1$ :  $K_p = \infty$ ,  $e(\infty) = 0$**

$$r(t) = A \cdot t$$

**Static velocity error constant**

$$K_v = \lim_{z \rightarrow 1} (z - 1) GH(z)$$

**Type 0:  $K_v = 0$ ,  $e(\infty) = \infty$**

**Type 1:  $K_v = \text{constant}$ ,**

**Type  $\geq 2$ :  $K_v = \infty$ ,  $e(\infty) = 0$**

$$r(t) = \frac{A}{2} t^2 \quad \text{Static acceleration error constant} \quad K_a = \lim_{z \rightarrow 1} (z - 1)^2 GH(z)$$

Type 0,1:  $K_a=0$ ,  $e(\infty)=\infty$

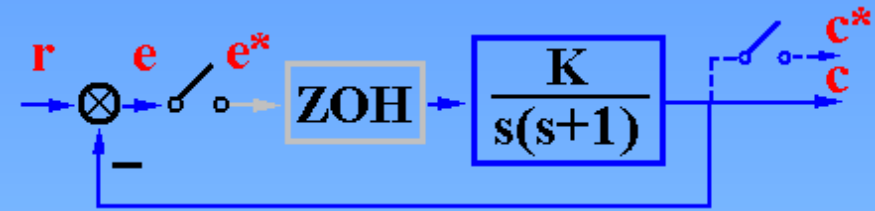
Type =2:  $K_a= \text{constant}$ ,

Type  $\geq 3$ :  $K_a= \infty$ ,  $e(\infty)=0$

$$\begin{cases} GH(z) = \frac{1}{(z-1)^v} GH_0(z) \\ \lim_{z \rightarrow 1} GH_0(z) = K \end{cases}$$

| 型別        | Static Error Constant |                         |                            | Steady-State Error                               |                                               |                                                     |
|-----------|-----------------------|-------------------------|----------------------------|--------------------------------------------------|-----------------------------------------------|-----------------------------------------------------|
| <b>V</b>  | $K_p = \lim GH(z)$    | $K_v = \lim (z-1)GH(z)$ | $K_a = \lim (z-1)^2 GH(z)$ | $r=A \cdot 1(t)$<br>$e(\infty) = -\frac{A}{K_p}$ | $r=A \cdot t$<br>$e(\infty) = \frac{AT}{K_v}$ | $r=A \cdot t^2/2$<br>$e(\infty) = \frac{AT^2}{K_a}$ |
| <b>0</b>  | $K_p$                 | 0                       | 0                          | $\frac{A}{K_p}$                                  | $\infty$                                      | $\infty$                                            |
| <b>I</b>  | $\infty$              | $K_v$                   | 0                          | 0                                                | $\frac{AT}{K_v}$                              | $\infty$                                            |
| <b>II</b> | $\infty$              | $\infty$                | $K_a$                      | 0                                                | 0                                             | $\frac{AT^2}{K_a}$                                  |

**Example 2** Consider the stable discrete system shown in the figure. When  $r(t)=2t$ , obtain  $e(\infty)$  with/without ZOH.



**Solution.**

no ZOH

$$\begin{cases} G(z) = Z\left[\frac{K}{s(s+1)}\right] = \frac{K(1-e^{-T})z}{(z-1)(z-e^{-T})} \\ K_v = \lim_{z \rightarrow 1} (z-1)G(z) = \lim_{z \rightarrow 1} \frac{K(1-e^{-T})z}{(z-e^{-T})} = K \end{cases}$$

$$e(\infty) = \frac{AT}{K_v} = \frac{2T}{K}$$

— dependent of T

with ZOH

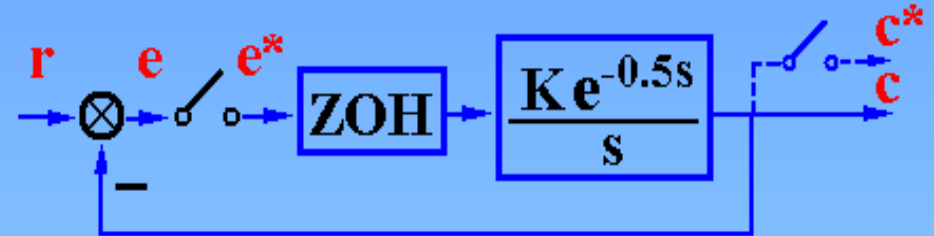
$$\begin{cases} G(z) = Z\left[\frac{1-e^{-Ts}}{s} \cdot \frac{K}{s(s+1)}\right] = K \frac{z-1}{z} \cdot Z\left[\frac{1}{s^2(s+1)}\right] \\ = K \frac{(T-1+e^{-T})z + (1-e^{-T}-Te^{-T})}{(z-1)(z-e^{-T})} \\ K_v = \lim_{z \rightarrow 1} (z-1)G(z) = \lim_{z \rightarrow 1} \frac{K(T-Te^{-T})}{z-e^{-T}} = KT \end{cases}$$

$$e(\infty) = \frac{AT}{K_v} = \frac{A}{K} = \frac{2}{K}$$

— independent of T

**Example 3** Consider the system shown in the figure,  $T=0.25$ . When  $r(t)=2\cdot 1(t)+t$ , obtain the range of  $K$  for  $e(\infty)<0.5$ .

**Solution.** The stable range of  $K$  is  
 $0 < K < 2.472$



$$G(z) = Z \left[ \frac{1 - e^{-Ts}}{s} \cdot \frac{K e^{-2Ts}}{s} \right]$$

$$= K(1 - z^{-1})z^{-2} Z \left[ \frac{1}{s^2} \right] = Kz^{-2} \frac{z-1}{z} \cdot \frac{Tz}{(z-1)^2} = \frac{KT}{z^2(z-1)} \quad v = 1$$

$$K_v = \lim_{z \rightarrow 1} (z-1)G(z) = \lim_{z \rightarrow 1} (z-1) \frac{KT}{z^2(z-1)} = KT$$

$$r_1(t) = 2 \cdot 1(t) \quad e_1(\infty) = 0$$

$$r_2(t) = t \quad e_2(\infty) = TA / K_v = 1 / K$$

$$2 < K < 2.472$$

$$e(\infty) = e_1(\infty) + e_2(\infty) = 1/K < 0.5 \Rightarrow K > 2$$

## 7.6.3 Steady-state error of discrete systems

(1) General method: obtain system response

(2) Final value theorem  $\left\{ \begin{array}{l} G(z) \rightarrow \Phi_e(z) \\ D(z) \rightarrow \text{Stability} \\ e(\infty) = \lim_{z \rightarrow 1} (z-1)R(z)\Phi_e(z) \end{array} \right.$

(3) Static error constant  $\left\{ \begin{array}{l} G(z) \rightarrow v, K_p, K_v, K_a \\ \text{Obtain } e(\infty) \end{array} \right.$